

Literature Review & Seminar

S. R. Mahadeva Prasanna

Professor, Dept. of DS & AI, IIIT Dharwad
Director, IIIT Dharwad

prasanna@iiitdwd.ac.in

January 1, 2025

Some Good Resources for LRS

- Chris Hart, “Doing a Literature Review - Releasing the research imagination” 2nd Edition, SAGE Publishers, Delhi 2018
- Arelene Fink, “Conducting Research Literature Reviews - From the internet to paper”, 5th Edition, SAGE Publishers, Delhi, 2020.
- Robert L. Jolles, “How to Run Seminars and Workshops”, Wiley, 2005
- Christine Harrington and Theresa Orosz, “Why the First-Year Seminar Matters: Helping Students Choose and Stay on a Career Path”, Rowman Littlefield Publishers, 2018.

LRS Course Planning

- 2 credit course ($1-0-0-4=2$)
- 28 contact hours
- $2 \times 14 = 28$ hrs
- $2 \times 2 = 4$ hrs by instructor
- $2 \times 12 = 24$ hrs by student seminars
- Wednesday, 6.30-8.30 pm
- 2-3 slots for each student in a semester

LRS Course Evaluation (1-0-0-4-2)

- Even though it is 2 credit course, going to be semester long course
- First instructional day (T_0)
- Except endsem, outstation candidates can take all other components of evaluation in online mode
- Midsem (15%) : As per calendar, $T_0 + 7W$;
- Endsem (25%): As per calendar (Offline for all), $T_0 + 14W$;
- Seminar Evaluation(40%): Continuous evaluation
- Literature Rev Report (20%): Continuous evaluation
- Attendance Policy : As per institute norms



Evaluation - contd

- More in action from students, seminar and report carries 60%
- Midsem (15%): Some questions from already held seminar and reports submitted, lit rev exercise ;
- Endsem (25%): Some questions from seminars and reports submitted, lit rev exercise ;

Seminars Evaluation

- 2-3 times seminar by each student
- Graded by course instructor, mentor/supervisor, committee of faculty members, entire batch of students
- Average score/grade will be announced after each seminar
- Final score / grade will be average of all seminars

Syllabus of LRS

- **Overview of Literature Review:** Introduction, searching for articles, reading and comprehending articles, summarizing works reported in the articles, critical review of the existing works and identifying gaps. (2 hrs)
- **Overview of Seminar:** Introduction, preparing for Seminar, art of good presentations, preparing a good set of slides, preparing a report, delivering seminars and question and answer sessions during seminar, measure of quality of seminar. (2 hrs)
- **Seminars by Enrolled Students:** Each student is expected to deliver 2-3 seminars during the course on the topic chosen by reviewing the literature and preparing slides. (20-22 hrs (2 hrs $\times$ 12 = 24 hrs of seminar slots + 64 hrs of self study for lit rev and preparing for seminars))

Syllabus of Intro to Research

- **Introduction:** Need, research vs course based study, qualifications, preparation - course work, comprehensive, etc.
- **Literature Review:** Searching, reading and comprehending, summarizing, critical review and identifying gaps.
- **Research Exploration:** Research questions, hypothesis, exploration, results and discussion.
- **Patenting and Publications:** Importance, steps, significance.
- **Presentation, Report and Thesis Writing:** Preparing, delivering and writing reports, thesis writing.
- **Conclusions and Future Scope:** Concluding, future directions.
- **Principles & Ethics in Research:** hard work, time management, health practices, patience, plagiarism .



Summary of Introduction to Research

- Why research?
- Preparing for doing research
- Course vs research based degree
- Good courses
- Literature review & seminar
- Research exploration
- Patents and publications
- Seminars
- Thesis writing and defence

Organization for Literature Review

- What is a literature review?
- Literature review vs academic research paper
- Need for writing a Literature Review
- Review vs Critical Review
- What to be done in Literature Review?
- Questions to ask during Literature Review
- Questions to ask while reading each research article
- What to write in a review article?
- How to organize a review article?

What is a literature review?

- A literature review discusses and analyses published information in a particular subject area.
- A literature review is more than a summary of the sources, it has an organizational pattern that combines both **summary and synthesis**.
- A **summary** is a recap of the important information of the source.
- A **synthesis** is a re-organization, or a reshuffling, of that information.
- It might give a new interpretation of old material or combine new with old interpretations.
- It might trace intellectual progression of field.
- Literature review may evaluate sources and advise reader on the most pertinent or relevant.

Literature review vs academic research paper

- Focus of an academic research paper is to document novelty and support your hypothesis or argument.
- Focus of a literature review is to summarize and synthesize arguments and ideas of others.
- Academic research paper also covers a range of sources, but only a select number of sources to support hypothesis or argument.
- Literature review can also have an argument, but it is not as important as covering a number of sources.
- Academic research paper and a literature review contain some of same elements.
- What aspect of study (either argument or sources) that is emphasized determines what type of document

Need for writing a Literature Review

- Literature review provide a handy guide to a particular topic.
- If you have limited time to conduct research, literature review can give you an overview or act as a stepping stone.
- For professionals, they are useful reports that keep them up to date with what is current in the field.
- For scholars, depth and breadth of literature review emphasizes credibility of writer in his or her field.
- Literature review also provides a solid background for a research paper's investigation.
- Comprehensive knowledge of the literature of the field is essential to identify gaps and hypothesize new directions.

Review vs Critical Review

- In literature review, purpose is to convey reader what knowledge and ideas have been established on a topic (**review**), and what their strengths and weaknesses are, accordingly, scope for future work (**critical review**).
- As a piece of writing, the literature review must be defined by a guiding concept.
- For instance, research objective, problem or issue planning to discuss, or argumentative thesis.
- It is not just a descriptive list of the material available, or a set of summaries

Merits of Literature Review

- Besides enlarging knowledge about the topic, writing a literature review lets gaining and demonstrating skills in two areas
- **Information seeking:** ability to scan literature efficiently, to identify a set of useful articles and books
- **Critical appraisal:** ability to apply principles of analysis to identify unbiased and valid studies.

What to be done in Literature Review?

- A literature review must be organized around and related directly to the thesis or research question you are developing
- Should synthesize results into a summary of what is and is not known
- Identify areas of gap in the literature
- Formulate questions that need further research

Questions to ask during Literature Review

- What is the research question that literature review helps to define?
- What is the scope of my literature review?
- What types of publications am I using (e.g., journals, conf, books)?
- How good was my information seeking? wide enough to ensure all the relevant material? narrow enough to exclude irrelevant material?
- Is number of sources used appropriate for the length of paper?
- Have I critically analysed the literature I use?
- Instead of just listing and summarizing items, discussing strengths and weaknesses?
- Have I cited and discussed studies contrary to my perspective?
- Will reader find literature review relevant, appropriate, and useful



Questions to ask while reading each research article

- Has the author formulated a problem/issue?
- Is it clearly defined? Is its significance clearly established?
- Could the problem have been approached from another perspective?
- Has the author evaluated the literature relevant to the problem?
- How accurate and valid are the measurements? Is analysis of data accurate and relevant to research question?
- Are conclusions validly based upon data and analysis?
- Is there an objective basis to reasoning, or is author merely “proving” what he or she already believes?
- In what ways this article contribute to understanding of problem under study? What are the strengths and limitations?



Before Starting Literature Review

- Roughly how many sources should you include?
- What types of sources (books, journal articles, websites)?
- Should you summarize, synthesize, or critique your sources by discussing a common theme or issue?
- Should you evaluate your sources?
- Should you provide subheadings and other background information, such as definitions and/or a history?

Reading some good Review Articles

- Look for other literature reviews in your area of interest or in the discipline and read them to get a sense of review article
- Will give ideas about types of themes you might want to look for in own research or ways to organize final review.
- You can simply put the word “review” in your search engine along with your other topic terms to find articles of this type on the Internet or in an electronic database.
- The bibliography or reference section of sources you’ve already read are also excellent entry points into your own research.

Narrow your Topic

- There are hundreds or even thousands of articles and books.
- The narrower the topic, the easier it will be to limit the number of sources need to read in order to get a good survey of the material.
- Your instructor will probably not expect you to read everything that's out there on the topic, but you'll make your job easier if you first limit your scope.
- And don't forget to tap into your professor's (or other professors') knowledge in the field.

What to write in a review article?

- Find a focus
- Convey it to your reader
- Consider organization
- Begin composing
- Revise, revise, revise
- Acknowledge works consulted

How to organize a review article?

- What is the most effective way of presenting the information?
- What are the most important topics, subtopics, etc., that review article needs to include?
- And in what order should you present them?
- Introduction, main body, critical review and discussion and conclusions/ recommendations.
- Main body: Chronological, trend, thematic, methodological

How Many Articles to Read and Summarize?

- No hard rule
- About 100 for PhD work (25 (5 + 10 + 10) for this course)
- About 50 for MTech(R) (15 (3 + 5 + 7) for this course)
- About 25 for UGR (8 for this course (2 + 3 + 3))

Organization for Seminar

- Introduction
- Research article vs presentation
- Need for presentation
- Significance of good presentation
- Research article vs report
- Need for report writing
- Significance of good reports

Introduction

- **Before** : Course work, comprehensive, literature review, research exploration, research articles/ patents.
- **Presentation**: Meant for Committee, non-specialists, etc. Lucid and highlighting only important achievements.
- **Report**: Ex. Synopsis Report. Meant for external examiners/ group / funding agency. Different organization compared to research article. But can significantly reuse materials from articles.
- **Thesis**: A novel that is connected from Title to last page. Writing story by taking help from literature review and published articles. Different than attaching articles. However can significantly reuse material from your published research article(s).

Research Article vs Presentation

- Specialists vs non-specialists
- Research arguments vs highlights
- Need not be lucid vs lucid
- Detailed vs gist
- Lengthy vs crispy
- Mostly text vs multimedia with audio-video illustrations

Need for Presentation

- Presenting research work to semi-specialists or non-specialists
- How to improve lucidity compared to research article?
- Multimedia way of presenting work
- Illustrations to drive home message
- Keep time factor in mind - 15 mt, 30 mt, 45 mt?
- Convincing audience about work done
- Follow some good presentation preparation skills

Significance of Good Presentation

- More clarifications / questions from audience
- Helps to spot missing link in the work from non-specialists view
- Suggestions for future work
- Audience react based on what you have projected in slides!
- More time on deciding what to put and what to eliminate
- Less text, more figures and tables, spread out, non-redundant - pleasing to audience

Research Article vs Report

- Specialist vs semi-specialist / non-specialist
- Argumentative vs reporting
- Motivation to sell novelty vs summative of work done and progress made
- Easy for specialist vs semi-specialist / non-specialist to read and get sense of work done

Need for Report

- Meant for bringing more visibility to group research activity, within and outside organization
- Meant for funding agencies to assess progress
- Meant for external examiners to get gist of work
- Demystifying and making information more readable from semi-specialists / non-specialist point of view

Significance of Good Report

- Captures all important findings in a summative form
- Relatively easier to follow
- Get measure of progress made
- Readers get an idea about novelty, quality and amount of work done!
- Helps in making material more oriented towards thesis writing
- It is like story writing as opposed to novel in case of PhD
- Follow some good report preparation skills

Thank You